

A critical re-reading of
Conformal field theory from lattice fermions
T. J. Osborne and A. Stottmeister, Commun. Math. Phys. **398**, 219–289 (2023);
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Abstract

This is an independent second pass over the technical core of [OS] (Sections 3.1–3.2, 4.2.3–4.2.4, 5, 6), carried out without deference to the earlier verification note in `osborne-stottmeister2023/`, which recorded “CORRECT, no logical gaps.” That verdict is too generous. The main theorems still look true, and I found no counterexample to any of them. But the paper contains **one invalid proof step that recurs in three places, one error estimate that is vacuous (indeed $+\infty$) as stated, a hierarchy of unstated regularity thresholds** whose numerical values are much larger than the folklore “ $K \geq 2$ suffices”, **one remark that does not follow from the lemma it invokes, a normalisation inconsistency that would change the central charge to $c = (L/\pi)^2$ if read literally, and two unproved ingredients that carry the whole proof of Theorem 6.3.** All numerical claims below were verified against explicitly constructed Daubechies filters; the scripts are in `numerics/`.

Summary of findings

#	Location	Issue	Severity
1	Lemma 4.5, p. 46	dominating function: the $\hat{s}$ -ratio bound is false	gap, repairable
2	(204)–(210), p. 53–54	Error ² (δ, L, k, N) = $+\infty$ for $k \neq 0$ (Ramond)	not established
3	“sufficiently regular”	three thresholds, none stated; $\delta=2$ needs $K \geq 9$	hypothesis gap
4	Remark 4.14, p. 50	$P^\pm$ does not preserve $\mathcal{D}_W$; claim does not follow	gap, repairable
5	Def. 4.2 (181)/(199) vs (165)/(167)	$\ell_{\pm,k}$ off by L/π ; (170) would give $c = (L/\pi)^2$	normalisation
6	Thm. 6.3 proof, p. 64	the two decisive ingredients are asserted, not proved	rigor debt
7	Thm. 4.16 preamble, p. 50	“preserving localization (Prop. 3.6)” not established	overstatement
8	(25), Lem. 3.7, (204)/(205)	typos / vacuous constants	cosmetic

Findings 1, 2 and 3 are one phenomenon seen from three sides, so I treat them together in Section 1–2. Nothing here contradicts Theorems A, B, C.

1 The $\hat{s}$ -ratio: a systematic invalid step

1.1 The zeros of $\hat{s}$

Throughout, s is a compactly supported orthonormal Daubechies scaling function of order K (so $2K$ taps), m_0 its low-pass symbol and $\hat{s}(\xi) = \prod_{j \geq 1} m_0(2^{-j}\xi)$ as in (102). Two elementary facts are used nowhere in [OS] but are decisive:

Observation 1. *Orthonormality gives $\sum_{n \in \mathbb{Z}} |\hat{s}(\xi + 2\pi n)|^2 = 1$; at $\xi = 0$ together with $\hat{s}(0) = 1$ this forces $\hat{s}(2\pi n) = 0$ for every $n \neq 0$. Moreover $\hat{s}(\xi) = (e^{-i\xi/2} \frac{\sin(\xi/2)}{\xi/2})^K \prod_{j \geq 1} L(2^{-j}\xi)$, so the zero at $2\pi n$ has order exactly K .*

Numerically (`numerics/os_check.py`, block [0]–[1]): $|\hat{s}(2\pi n)| \leq 10^{-12}$ for db2, ..., db6 and $n = 1, 2, 3, 5$, and $|\hat{s}(2\pi + \delta)| \sim \delta^p$ with $p = 2.00, 2.18, 3.08$ for $K = 2, 3, 4$.

1.2 Finding 1: the dominating function in Lemma 4.5

Finding 1. The domination step in the proof of Lemma 4.5 is invalid, and the stated inequality is false.

The proof reduces convergence to $\ell^2(\Gamma_{\infty, \pm}, (2L)^{-1})$ -convergence of

$$f_k^{(N)}(M, m) = \hat{s}(\varepsilon_N m) \cos(\tfrac{1}{4}\varepsilon_N k)^2 \frac{\sin(\varepsilon_N(m + \frac{k}{2}))}{\varepsilon_N} \prod_{j=1}^{N-M} m_0(\varepsilon_{M+j}(m+k)) \longrightarrow f_k(M, m) = \hat{s}(\varepsilon_M(m+k)) (m + \frac{k}{2})$$

and asserts the existence of an ℓ^2 -dominating function via

$$|f_k^{(N)}(M, m)| \leq \left| (m + \frac{k}{2}) \hat{s}(\varepsilon_M(m+k)) \right| \left| \frac{\hat{s}(\varepsilon_N m)}{\hat{s}(\varepsilon_N(m+k))} \right| \leq C_k \left| (m + \frac{k}{2}) \hat{s}(\varepsilon_M(m+k)) \right|, \quad (1)$$

“for some $C_k > 0$, because of the uniform continuity of $\hat{s}$.”

Two things go wrong. First, *uniform continuity of $\hat{s}$ controls the difference $\hat{s}(\varepsilon_N m) - \hat{s}(\varepsilon_N(m+k))$, never the quotient*; bounding a quotient needs a lower bound on $|\hat{s}|$, which is exactly what Observation 1 denies. Second, and concretely: in the Ramond sector $\Gamma_{\infty, +} = \frac{\pi}{L}\mathbb{Z}$ one may choose

$$m + k = \frac{2\pi n}{\varepsilon_N} = \frac{\pi}{L} 2^{N+1} n \in \Gamma_{\infty, +}, \quad n \neq 0,$$

so that $\hat{s}(\varepsilon_N(m+k)) = \hat{s}(2\pi n) = 0$ while $\hat{s}(\varepsilon_N m) = \hat{s}(2\pi n - \varepsilon_N k) \neq 0$ for $k \neq 0$. The middle expression in (1) is then ∞ , and the right-hand side is 0 (because $\hat{s}(\varepsilon_M(m+k)) = \hat{s}(2^{N-M} 2\pi n) = 0$ as well), while the left-hand side is strictly positive. *No constant C_k makes (1) true.*

Numerically (`os_check2.py`, block [4]), for db4, $M = 2$, $L = \pi$:

N	k	m	$ f_k^{(N)}(M, m) $	claimed bound $C_k f_k(M, m) $
6	2	126	$1.547 \cdot 10^{-6}$	$C_k \cdot 7.7 \cdot 10^{-27}$ (i.e. 0)
8	2	510	$5.936 \cdot 10^{-9}$	$C_k \cdot 3.1 \cdot 10^{-26}$ (i.e. 0)
10	2	2046	$2.307 \cdot 10^{-11}$	$C_k \cdot 1.2 \cdot 10^{-25}$ (i.e. 0)

(The right-hand column is machine zero: $\hat{s}$ vanishes there to order K .)

1.3 Why the lemma is nevertheless true, and how to repair the proof

The saving grace is that at these resonances the *sine* factor is also near a zero: $\varepsilon_N(m + \frac{k}{2}) = 2\pi n - \frac{1}{2}\varepsilon_N k$, so $\sin(\varepsilon_N(m + \frac{k}{2}))/\varepsilon_N \approx -k/2$ rather than $\approx m$. Together with $|\hat{s}(2\pi n - \varepsilon_N k)| \asymp c_n(\varepsilon_N |k|)^K$ this gives $|f_k^{(N)}(M, m)| = O(\varepsilon_N^K)$ at the resonance points. Measured decay (`os_check2.py`, block [6]): 2^{-pN} with $p = 2.003, 3.003, 4.005$ for db2, 3, 4 — i.e. exactly $p = K$.

Claim 2 (repair). *Do not form the quotient. Bound instead*

$$|f_k^{(N)}(M, m)| \leq \left| m + \frac{k}{2} \right| |\hat{s}(\varepsilon_N m)| \prod_{j=1}^{N-M} |m_0(\varepsilon_{M+j}(m+k))|,$$

and estimate the finite product directly by the paper’s own Lemma 3.7 inside the N -th Brillouin zone ($|m+k| \leq \pi/\varepsilon_N$), and by $|\hat{s}(\varepsilon_N m)| \leq C(1 + \varepsilon_N |m|)^{-(K-\bar{K})}$ together with the $2^{N-M+1}\pi$ -periodicity of the product outside it. This yields $g_k(M, m) = C_{k,M}(1 + |m|)^{-(K-\bar{K}-1)}$, which lies in ℓ^2 under the same condition as the paper’s own g_k .

Lemma 3.7 is proved in [OS] for precisely this purpose (“a decay estimate on the finite products ... that directly implies (102) ... by dominated convergence”), and is then not used in the proof of Lemma 4.5. The repair is therefore book-keeping, not new mathematics.

Two numerical confirmations (`os_check2.py` [5], `os_check3.py` [9]):

- $\sum_m |f_k^{(N)}(M, m) - f_k(M, m)|^2 \rightarrow 0$ monotonically; for db4, $M = 2$, $k = 2$ it falls from $1.36 \cdot 10^1$ at $N = 4$ to $1.01 \cdot 10^{-4}$ at $N = 14$, with a stable ratio ≈ 0.33 per step. *The conclusion of Lemma 4.5 is correct.*
- $G(m) := \sup_{N > M} |f_k^{(N)}(M, m)|$ has the *same* decay exponent as $|f_k(M, m)|$ on every dyadic window up to $m = 2^{13}$, and $\sum_m G(m)^2 / \sum_m |f_k(M, m)|^2 = 1.008$. A valid dominating function exists; it is just not the one exhibited.

1.4 Finding 2: the error estimates (204)–(210) inherit the singularity

Finding 2. $\text{Error}^2(\delta, L, k, N)$ as defined in (206) equals $+\infty$ for every $k \neq 0$ and every N in the Ramond sector. The passage from (208), line 1, to (208), line 2, is vacuous for $k \neq 0$ in both sectors. Consequently (210) — and with it (207) — is established only for $k = 0$.

The factorisation (204)–(205) pulls $|(m + \frac{k}{2})\hat{s}(\varepsilon_M(m + k))|^2$ out of the sum and leaves behind the same quotient $\hat{s}(\varepsilon_N m) / \hat{s}(\varepsilon_N(m + k))$, now inside a supremum. In the Ramond sector the supremum runs over a set containing an exact pole (same m as above), so $\text{Error}^2 = +\infty$. In (208), line 2, the discrete supremum over $m \in \Gamma_{\infty, \pm}$ is relaxed to $\sup_{x \in \mathbb{R}}$; since $x \mapsto x + \varepsilon_N k$ can be placed on a zero of $\hat{s}$ while x is not, that supremum is $+\infty$ for every $k \neq 0$ in both sectors. The step from (209) to (210) is then carried entirely by the sentence “as long as N is sufficiently large, we expect essentially the same behavior for $k \neq 0$ because of the uniform continuity of $\hat{s}$ ”, which is the invalid step of Finding 1 in supremum form.

Measured (`os_check2.py` [7], `os_check3.py` [10]), db4, $\delta = 2$:

	$N = 6$	$N = 8$	$N = 10$	$N = 12$
Error^2 , Ramond, $k = 1$	$+\infty$	$+\infty$	$\geq 3.9 \cdot 10^{12}$	—
Error^2 , NS, $k = 1$	$1.21 \cdot 10^{-7}$	$5.61 \cdot 10^{-10}$	$2.35 \cdot 10^{-12}$	$9.44 \cdot 10^{-15}$
successive ratios (NS)	215, 239, 249	vs. $2^{2\delta \cdot 2} = 256$ predicted by (210)		

So the *conclusion* (210) is numerically correct in the Neveu–Schwarz sector, false in the Ramond sector, and derived by an invalid argument in both. Since the paper’s own footnote 15 already flags that the Ramond sector needs extra care with χ_{Γ_N} , this is a natural place for the fix.

What is unaffected — and it is the part that matters downstream. Remark 4.21 / eq. (212), the $k = 0$ time-evolution bound

$$\|(\ell_{\pm, 0}^{(N)} - \ell_{\pm, 0})_{\pm\pm} R_{\infty}^M(\eta_p)\|_{\infty}^2 \leq \varepsilon_N^{2\delta} \left(\sup_x \frac{|\text{sinc}(x) - 1|^2}{|x|^{2\delta}} \right) \|\eta\|_M^2 \left(\frac{L}{\pi}\right)^2 \sum_{n \in \Gamma_{M, \pm}} \theta(\pm n) |n|^{2(1+\delta)},$$

comes from the *momentum-cutoff* route (201) and contains no $\hat{s}$ at all: the sum is finite (over the M -th Brillouin zone) and no regularity of s enters. I confirmed $\sup_x |\text{sinc}(x) - 1|^2 / |x|^4 = 1/36$, attained as $x \rightarrow 0$ (it follows from $1 - x^2/6 \leq \text{sinc}(x) \leq 1$). *The advertised $\sim 2^{-2N}$ error scaling for the time evolution is sound and unconditional.* This is the estimate that feeds the companion quantum-simulation paper; the $k \neq 0$ budget from (210) is not.

2 Finding 3: what “sufficiently regular” actually costs

Finding 3. The hypothesis “ $s \in C^\alpha(\mathbb{R})$ sufficiently regular” is never quantified, and carries *three different* thresholds. In terms of the Sobolev exponent σ_K of the Daubechies- K scaling function:

- (a) $\mathcal{D}_W \subset \mathcal{D}(\ell_{\pm,k})$ (needed for $\mathcal{D}_W$ to be a core at all, p. 44) and ℓ^2 -domination in Lemma 4.5 both require $\sigma_K > 1$;
- (b) the two Cauchy–Schwarz splittings in the proof of Lemma 4.15 require $\delta > 3/2$ (second factor, since $\|\tilde{\ell}_{\pm,k}^{(N)} R_\infty^K(e_l)\|$ grows linearly in k) and then $\sigma_K > \delta$ (first factor, since $\hat{X}_{\text{per}}$ is periodic and does not decay) — hence $\sigma_K > 3/2$;
- (c) finiteness of $\|\hat{s}(\varepsilon_M(\cdot + \frac{k}{2}))\|_{h^{1+\delta}}$ in (207) requires $\sigma_K > 1 + \delta$; for the $\delta = 2$ used to advertise the $2^{-2\delta N}$ rate this is $\sigma_K > 3$.

I constructed the Daubechies filters from scratch (spectral factorisation of $\cos^{2K}(\xi/2)P_K(\sin^2(\xi/2))$; orthonormality verified to $< 6 \cdot 10^{-15}$) and measured σ_K from dyadic-block L^2 decay of $\hat{s}$; the values reproduce the standard Villemoes/Daubechies table to 3–4 digits (`os_check4.py`).

K	2	3	4	5	6	7	8	9	10	11	12
σ_K (measured)	1.002	1.416	1.776	2.098	2.390	2.661	2.917	3.165	3.406	3.641	3.873
$K - \bar{K}$ (sup-envelope)	1.336	1.634	1.910	2.174	2.431	2.681	2.927	3.170	3.409	3.646	3.881
(a) $\sigma_K > 1$	≈	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
(b) $\sigma_K > 3/2$	×	×	✓	✓	✓	✓	✓	✓	✓	✓	✓
(c) $\sigma_K > 3$	×	×	×	×	×	×	×	✓	✓	✓	✓

Three consequences worth stating in a revision:

- db2 is *exactly borderline* ($\sigma_2 = 1$, so $s \notin H^1$): the wavelet core $\mathcal{D}_W$ is then not contained in $\mathcal{D}(\ell_{\pm,k})$ and Lemma 4.5 has no content. The honest minimum for Section 4.2.4 is $K \geq 3$, and $K \geq 4$ once Lemma 4.15 is included.
- The advertised $\delta = 2$ rate in (210) requires $K \geq 9$ — *not* $K \geq 2$, and not the “ $K \geq 6$ suffices” recorded in the earlier verification note in `osborne-stottmeister2023/`. Since $\text{supp}(Ks) = [0, 2K - 1]$, Remark 4.20’s own constraint $\varepsilon_M(2K - 1) \sim L$ then forces a correspondingly coarser M : the regularity needed for the rate and the localisation needed for the qubit count pull against each other, and the paper does not resolve the trade-off.
- The relevant exponent differs between the two uses: the pointwise bound (109) supplies $K - \bar{K}$, whereas the ℓ^2 -sums actually need the (larger) Sobolev exponent. Using (109) to check the sums, as the paper does, is lossy by roughly 0.4–0.5 in K .

3 Finding 4: Remark 4.14 does not follow from Lemma 4.5

Remark 4.14 states that in the NS sector the Hilbert–Schmidt norms in (201) vanish for $k = 0, \pm\pi/L$ (the Moebius generators), and concludes that “*a simple extension of Lemma 4.5 is sufficient to prove Theorem 4.11 in this case using the wavelet renormalization group.*”

The vanishing is correct, though for a subtler reason than emptiness of the index set: for $k = -\pi/L$ and the $+$ chirality the constraint $\theta(n - k)\theta(-n)$ admits $n = -\pi/2L \in \Gamma_{\infty,-}$, and the summand vanishes there because $n - \frac{k}{2} = 0$ kills both $\sin(\varepsilon_N(n - \frac{k}{2}))$ and $(n - \frac{k}{2})$.

Finding 4. The conclusion does not follow. Estimate (200) also requires the *diagonal* blocks $(\tilde{o}^{(N)} - o)_{\pm\pm} = S_\pm(\tilde{o}^{(N)} - o)S_\pm$ applied to $R_\infty^M(\eta)$, i.e. convergence on the Hardy-projected vectors $P^\mp \xi$. But $P^\pm$ does not map $\mathcal{D}_W$ into $\mathcal{D}_W$.

$\mathcal{D}_W = \bigcup_N R_\infty^N(\mathfrak{h}_{N,\pm}) = \bigcup_N V_N$ is the union of the multiresolution spaces; a V_M element has $\hat{\xi}(k) = \varepsilon_M^{1/2} \hat{s}(\varepsilon_M k) \hat{\eta}_{\text{per}}(k)$ with $\hat{\eta}_{\text{per}}$ periodic, so it cannot be supported on positive frequencies alone without vanishing. Measured relative distance of $P^+_{s(\varepsilon_M)}$ from V_J (`os_check3.py` [11], db4, $M = 3$): 0.353, 0.080, 0.0074, $6.4 \cdot 10^{-4}$, $5.5 \cdot 10^{-5}$ for $J = 3, 4, 6, 8, 10$ — convergent to 0, as it must be by density, but never 0.

Claim 3 (repair). $\ell_{\pm,k}$ and $\tilde{\ell}_{\pm,k}^{(N)}$ are bounded $h^1(\Gamma_{\infty,\pm}) \rightarrow h^0(\Gamma_{\infty,\pm})$ uniformly in N (their symbols are $O(|m|)$, and $(R_\infty^N)^*$ is bounded $h^1 \rightarrow h^1(\Gamma_N)$ by the decay of $\hat{s}$). Since $\mathcal{D}_W$ is h^1 -dense (the paper’s own graph-norm argument, p. 44) and $P^\pm$ is a Fourier multiplier preserving h^1 , strong convergence extends from $\mathcal{D}_W$ to all of $h^1 \ni P^\mp \xi$. This closes the remark, but it is not “a simple extension of Lemma 4.5”.

This also isolates an unremarked structural reason why the momentum-cutoff RG is needed: in that setting $P^\pm$ is diagonal in the plane-wave basis, so $\mathcal{D}_{\text{std}}$ is invariant and (200) applies verbatim. The paper motivates the switch by the Hilbert–Schmidt condition alone.

4 Finding 5: a normalisation inconsistency in Definition 4.2

Two incompatible normalisations of the one-particle Virasoro generator are in circulation:

$$(A) \text{ eq. (143)/(165): } \ell_k(l) = -(l + \frac{k}{2})p_-, \quad (B) \text{ Def. 4.2 (181) + (199): } \ell_{\pm,k}e_n = \pm \frac{L}{\pi}(n \mp \frac{k}{2})e_{n \mp k}.$$

(B) is internally consistent — (181), (184), (198), (199) and the estimates (200)/(201) all use it, and $\tilde{\ell}_{\pm,k}^{(N)} \rightarrow \ell_{\pm,k}$ holds in (B) — but it is (A) that is compatible with the bridge formula $L_{\pm,k} = \frac{L}{\pi} dF_S(\ell_{\pm,k})$ used in (142), (145), (167), (170) and (194). The two differ by exactly L/π .

Finding 5. Read literally with Definition 4.2, the commutator (170) acquires the structure constant $(L/\pi)^2(k - k')$ instead of $\frac{L}{\pi}(k - k')$, and its central term becomes $(L/\pi)^2 \cdot \frac{1}{12}n(n^2 - 1)$ with $n = \frac{L}{\pi}k$, i.e. $c = (L/\pi)^2$ rather than $c = 1$. The same L/π propagates into (173), (177), (178) and (194).

Direct evaluation of the Schwinger cocycle $c_{P^\mp}(\ell_{+,k}, \ell_{+,-k})$ over the finite strip $0 < |l| < |k|$ (`os_check5.py`), target $\frac{1}{12}n(n^2 - 1)$:

L	n	$(L/\pi)^2 c_S$ with (A)	$(L/\pi)^2 c_S$ with (B)
π	3	2.000	2.000
2π	3	2.000	8.000
5	3	2.000	5.066
2π	5	10.000	40.000

The two agree iff $L = \pi$, which the paper nowhere assumes (L is free throughout; $\varepsilon_N L_N = L$). *No convergence statement is affected* — the factor is common to approximant and limit — but the identification of the limit with the $c = 1$ (resp. $c = \frac{1}{2}$) Virasoro generators requires normalisation (A), so Definition 4.2 should read $\ell_{\pm,k} = \pm \frac{1}{2L} \sum_l (l \mp \frac{k}{2}) e_{l \mp k} \otimes e_l$ (equivalently, drop L/π from (199)).

5 Finding 6: Theorem 6.3 rests on two unproved assertions

The telescoping estimates (236)/(237) reduce the statement to applying Theorem 4.16 to the vectors $\prod_{p < q} L_\pm(X_p)\Omega_0$. The proof then closes with:

“First, $\|\prod_{p=1}^n : (\pi_\pm(\alpha_\infty^N(L_\pm^{(N)}(X_{N,p}))) : \Omega_0\|$ can be bounded uniformly in N ... Second, $\prod_{p=1}^n L_\pm(X_p)\Omega_0$ is given by a convergent sum ... of Fock-space vectors with finite particle number in momentum space.”

Finding 6. These two “observations” are exactly the decisive ingredients and neither is proved.

They are not decorative. Theorem 4.16 gives strong convergence only on $\mathfrak{F}_a^{\text{alg}}(\mathcal{D}_{\text{std}})$, and the intermediate vectors are *not* in that domain. Note the contrast with the unsmeared case: for fixed k the off-diagonal block $(\ell_{\pm,k})_{\pm\mp}$ is supported on the finite strip $0 < |n| < |k|$, hence *finite rank*, so $L_{\pm,k}\Omega_0$ is a finite combination of plane-wave pairs. After smearing, $L_{\pm}(X) = \frac{1}{2L} \sum_k \hat{X}_k L_{\pm,k}$ has an off-diagonal block that is Hilbert–Schmidt but of infinite rank, so $L_{\pm}(X)\Omega_0 = a^\dagger G_{+-} a^\dagger \Omega_0$ is an infinite two-particle superposition, outside $\mathfrak{F}_a^{\text{alg}}(\mathcal{D}_{\text{std}})$ and of unbounded $L_{\pm,0}$ -energy.

What is needed is an *N-uniform energy bound* $\| : \pi_{\pm}(\alpha_{\infty}^N(L_{\pm}^{(N)}(X))) : \Psi \| \leq C_X \|(1 + L_{\pm,0})\Psi\|$, the lattice analogue of the Buchholz–Schulz–Mirbach / Carpi–Weiner energy bounds for smeared Virasoro fields. That is a theorem in its own right, and it is the main rigor debt of Section 6. The same gap does not affect Theorem 6.1, where the insertions are bounded elements of the CAR algebra and the argument is complete.

Two smaller items in the same vicinity. Theorem 4.11 is stated with no proof at all (it is offered as an immediate consequence of Lemma 4.10 and (200)); and the assertion that $\mathfrak{F}_a^{\text{alg}}(\mathcal{D}_{\text{std}})$ contains a total set of analytic vectors for $L_{\pm,k}$ is justified only by a reference to Corollary 4.3 and (200).

That assertion does not follow, and it cannot be had by iterating (25). In $\pi_{\pm}$ the pair terms $a^\dagger G_{+-} a^\dagger$ and $a G_{-+} a$ raise the particle number, so at step j the estimate is applied at level $n + 2j$ rather than at level n , and iterating (25) together with (187) produces a factor $(j!)^2$ after j steps; the exponential series then diverges for every $t > 0$.¹ The statement itself is presumably true, being the Goodman–Wallach/Carpi–Weiner analyticity of finite-energy vectors, but it does not follow from the estimates available here.

What repairs Theorem 4.11 is therefore not an analytic-vector argument at all, but Nelson’s *commutator* theorem [Reed–Simon X.37] applied with $N = 1 + L_{\pm,0}$: its first hypothesis is exactly the uniform energy bound demanded above, and its second follows from the Virasoro relation $[L_{\pm,0}, L_{\pm,k}] = -\frac{L}{\pi} k L_{\pm,k}$ together with the form bound. The conclusion is stronger than the one claimed — essential self-adjointness on *every* core for $1 + L_{\pm,0}$, in particular on $\mathcal{D}(L_{\pm,0})$.

6 Finding 7: the locality claim for Theorem 4.16

The preamble to Lemma 4.15 states that combining the two renormalisation groups exploits convergence in $\pi_{\pm}$ “while, at the same time, preserving localization in real space in the sense of Proposition 3.6.”

Finding 7. This is not established. Proposition 3.6 is a statement about the *wavelet* RG. In Theorem 4.16 the fermion algebra is transported by the *momentum-cutoff* maps α_{∞}^N , which by the paper’s own eq. (112) send lattice-localised operators to operators that are not localised. The wavelet RG enters only through the smearing functions $S_N^M(X)$, and inspection of the proof of Lemma 4.15 shows its role there is summability of the k -sum (the weights $(1 + |\varepsilon_M k|)^{\pm 2\delta}$), not locality.

Compounding this, the approximants used in Theorem 4.16 are the *modified* ones (197), whose real-space form (159)–(162) is a convolution with $\widehat{\chi_{\Gamma_N}}$ — a Dirichlet kernel with only $O(1/|x|)$ decay. What is true, and worth saying instead, is that the smearing function is wavelet-localised at scale M ; the observable is quasi-local only up to Dirichlet tails. The distinction matters for the quantum-simulation reading of these results, where locality of the simulated operator is the whole point.

¹The first version of this note asserted the opposite, deriving a radius $t < 1/(2\|G_{+-}\|_2)$ from $\|(a^\dagger G_{+-} a^\dagger)^j P_{\leq n}\| \leq 2^j j! j^{n/2} \|G_{+-}\|_2^j$. That bound holds the particle number fixed at n and is wrong.

7 Finding 8: minor corrigenda

1. **(25), p. 15.** “ $\|aG_{+-}aP_{\leq n}\| \leq n\|G_{+-}\|_2$ ” should read $\|aG_{+-}aP_{\leq n}\| \leq n\|G_{+-}\|_2$ — which is how it is used in (200) and in (22)/(24).
2. **Lemma 3.7, p. 27.** The display asserts “=” where “ $\leq$ ” is meant. Also $\max\{e^C, \pi^{-K_j}\} = e^C$ identically, since $q_j \geq 1$ forces $K_j \geq 0$; and case 3 of the proof claims $2^{MK_j} \leq \pi^{-K_j}(1+|l|)^{K_j}$, which fails throughout $2^M < |l| < \pi 2^M - 1$. The correct step is simply $2^{MK_j} \leq (1+|l|)^{K_j}$, i.e. drop π^{-K_j} .
3. **(204)/(205), p. 53.** The factorisation divides by $(m + \frac{k}{2})$, which vanishes at $m = -k/2$ whenever $k \in \frac{2\pi}{L}\mathbb{Z}$ (removable, since $\sin$ vanishes there too). The Sobolev norm in (205)–(207) is written with $\hat{s}(\varepsilon_M(\cdot + \frac{k}{2}))$ while (204) carries $\hat{s}(\varepsilon_M(\cdot + k))$.
4. **(213), p. 56.** From $\binom{2j}{j}^{1/2} \leq 2^j$ and $\prod_{i=0}^{j-1}(a+ib) \leq a^j(2j-1)!! \leq a^j 2^j j!$ with $a = |m| + \frac{1}{2}|k|$, $b = |k| \leq 2a$, the bound is $(4a)^j j!$, not $(2a)^j j!$; (213) should then carry $\frac{2L}{\pi}(|m| + \frac{1}{2}|k|)t$ in place of $\frac{L}{\pi}(|m| + \frac{1}{2}|k|)t$. The conclusion (a positive radius of convergence) is unchanged.
5. **§4.2.2.** The lattice Virasoro relation (146) and the $O(\varepsilon_N^2)$ remainders (147) are computed for the *unmodified* approximants; the modified ones (160)/(162) carrying χ_{Γ_N} — which are the ones used in Theorem 4.11 and 4.16 — are only argued to agree asymptotically. Their finite- N algebra is not checked. Harmless for the theorems, but the sentence “we recover the original Koo–Saleur formulas for $N \rightarrow \infty$ ” is doing more work than it looks.

Verdict

Theorems A, B and C (= 4.11, 6.1, 6.3, with 4.6, 4.16, 5.1) are, as far as I can tell, true, and the architecture — one-particle $\rightarrow$ Fock $\rightarrow$ correlation functions, with two renormalisation groups doing two different jobs — is sound and, in my view, the right way to do this. But the earlier “no logical gaps” verdict does not survive contact with the details. Ranked by what a referee should have caught:

1. **The $\hat{s}$ -quotient (Findings 1–2).** An invalid step, used three times, that is *false* rather than merely unjustified, and that makes the advertised $k \neq 0$ error estimate infinite in the Ramond sector. Lemma 4.5 is repairable in a paragraph using the paper’s own Lemma 3.7; (210) needs genuine rewriting for $k \neq 0$.
2. **Theorem 6.3 (Finding 6).** Two unproved assertions carry the proof, and one of them (an N -uniform energy bound) is a real theorem.
3. **Regularity (Finding 3).** “Sufficiently regular” hides $K \geq 3$, $K \geq 4$ and $K \geq 9$ in three different places. The $K \geq 9$ is the interesting one: the paper’s headline practical claim — explicit error bounds for quantum simulation — is stated at $\delta = 2$, which no Daubechies wavelet below order 9 supports along the wavelet route.
4. **Findings 4, 5, 7.** Each is a paragraph-sized fix, but 5 would read as a factual error about the central charge to anyone who takes Definition 4.2 at face value.

Net: the paper is right, and it is under-proved in a way that a careful revision could fix in about three pages. The one thing I would not simply patch is (210): the honest statement is that the wavelet route gives an explicit rate only for $k = 0$, and that the clean, unconditional error bound available today is the momentum-cutoff estimate (212).

A Numerical protocol

All checks are in `lattice_cft/numerics/` (pure `numpy`, no external wavelet library). Conventions: $L = \pi$, so $\Gamma_{\infty,+} = \mathbb{Z}$, $\Gamma_{\infty,-} = \mathbb{Z} + \frac{1}{2}$, $\varepsilon_N = 2^{-N}\pi$; $m_0(\xi) = 2^{-1/2} \sum_n h_n e^{-in\xi}$, $\hat{s}(\xi) = \prod_{j \geq 1} m_0(2^{-j}\xi)$ truncated adaptively ($J = \lceil \log_2 \max |\xi| \rceil + 32$, the tail being $1 + O(2^{-J}|\xi|)$).

- `os_check.py` — filter sanity ($\sum h_n = \sqrt{2}$, $\langle h, h(\cdot - 2k) \rangle = \delta$ to $< 10^{-12}$), order of the zeros of $\hat{s}$ at $2\pi n$, the $\hat{s}$ -quotient in both sectors.
- `os_check2.py` — cross-check of the product form against the quotient form of $f_k^{(N)}$ ($< 2 \cdot 10^{-14}$ where both are defined); failure of the stated dominating function; ℓ^2 -convergence of $f_k^{(N)} \rightarrow f_k$; 2^{-KN} decay of the resonance spikes; Error² in the Ramond sector.
- `os_check3.py` — decay exponents; existence of a valid dominating function; Error² in the NS sector; distance of $P^+_{\mathcal{S}(\varepsilon_M)}$ from V_J .
- `os_check4.py` — Daubechies filters for $K = 2, \dots, 12$ by spectral factorisation (orthonormality $< 6 \cdot 10^{-15}$); Sobolev exponents σ_K from dyadic-block L^2 decay, reproducing the literature values ($\sigma_2 = 1.000$, $\sigma_3 = 1.415$, $\sigma_4 = 1.776$, $\sigma_6 = 2.386$, $\sigma_8 = 2.917$) to 3–4 digits.
- `os_check5.py` — Schwinger cocycle over the finite strip, both normalisations.